market_exposed_legacy

equity strategy | family: equity-directional | status: reference | derived from `market_exposed`

What this strategy does

Balanced Leverage signal, longs and shorts not forced to match, market moves left in -- ORIGINAL leverage cap (new positions only, open book ignored)

Why it should work. Let net exposure float with the signal; empirically this smuggles in uncontrolled market beta.

Design notes (from the strategy's source file).

Market Exposed -- like Balanced Leverage, but longs and shorts need not match, so it rides the market (cautionary result).

Balanced Leverage's signal, but the long and short legs are no longer forced to equal dollar totals, and returns are RAW so real market beta is carried. Net dollar exposure is only ~10% of gross (as in every other book) yet the book is 0.52-correlated with the market, because raw returns carry each stock's own beta and equal dollars long and short are not equal BETA: high-SUE and low-SUE firms differ in beta. Kept as a documented counter-example to the explicit beta overlay.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	raw
cap_mode	priority
exposure_cap	entry_cohort
beta_overlay	0.0

Performance

metric	market exposed legacy	market (CRSP VW)
Annualized return	7.30%	9.94%
Annualized volatility	6.56%	19.35%
Sharpe (NAV-pct)	1.11	0.59
Sortino	1.61	
Max drawdown	-15.5%	-55.5%
Calmar	0.47	
Final NAV ($10M start)	$39,420,102	
Worst day	-3.54%	
% positive days	51.7%	
Sharpe 1996-2001	1.79	
Sharpe 2002-2007	1.48	

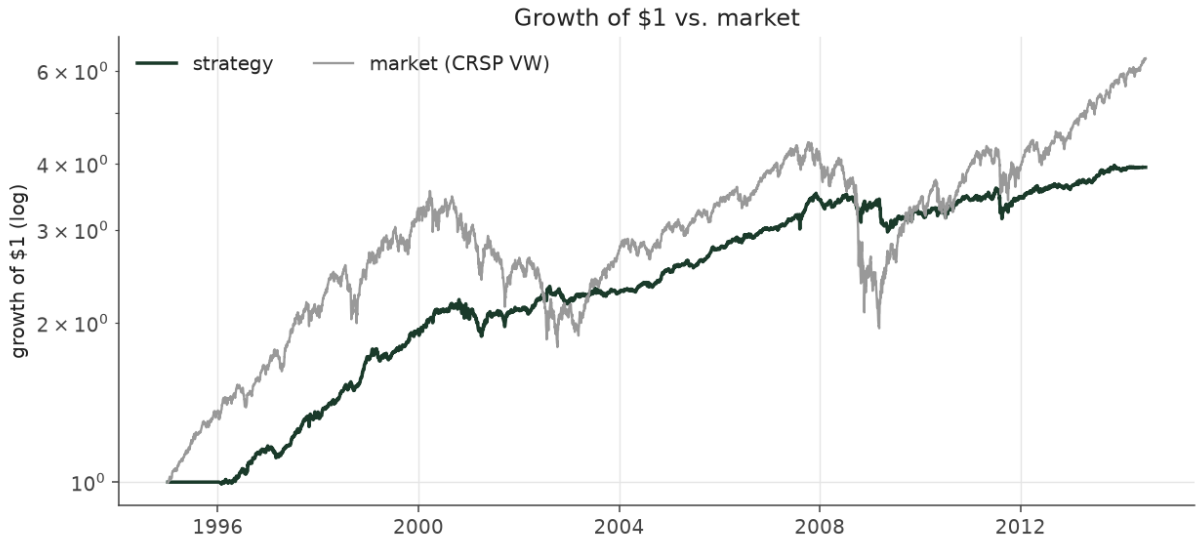
Sharpe 2008-2013	0.32	

Correlation of daily returns with the market: **0.52**.

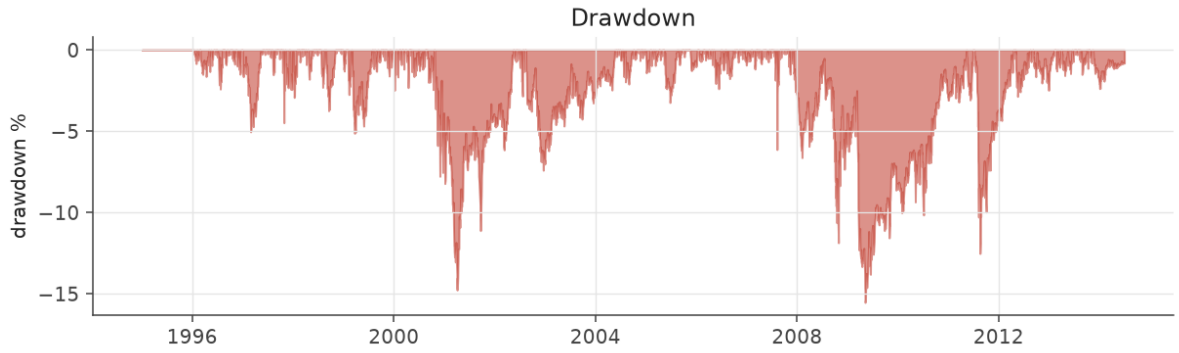
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.11

• best return among equity: `market_boost_max` (0.116); this one: 0.073

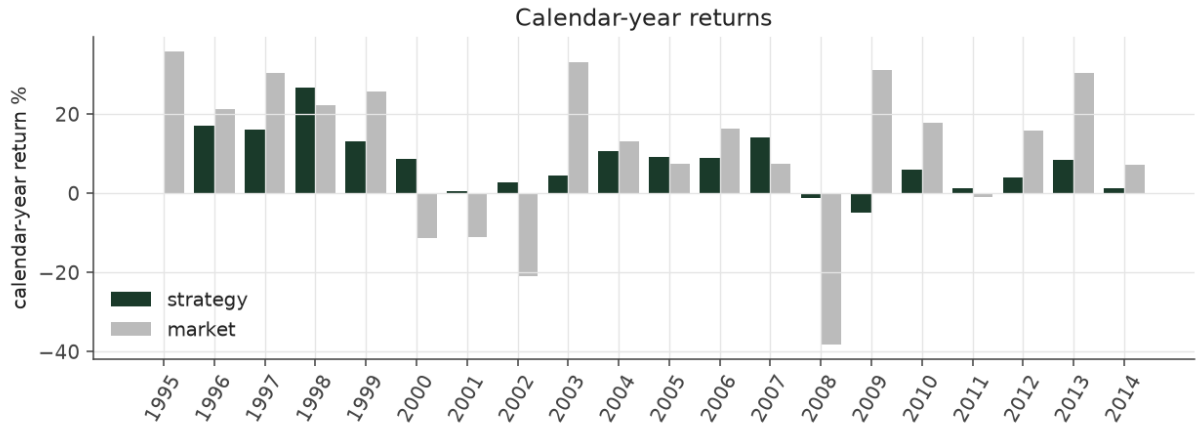
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.47



NAV (log scale)



Drawdown from running peak

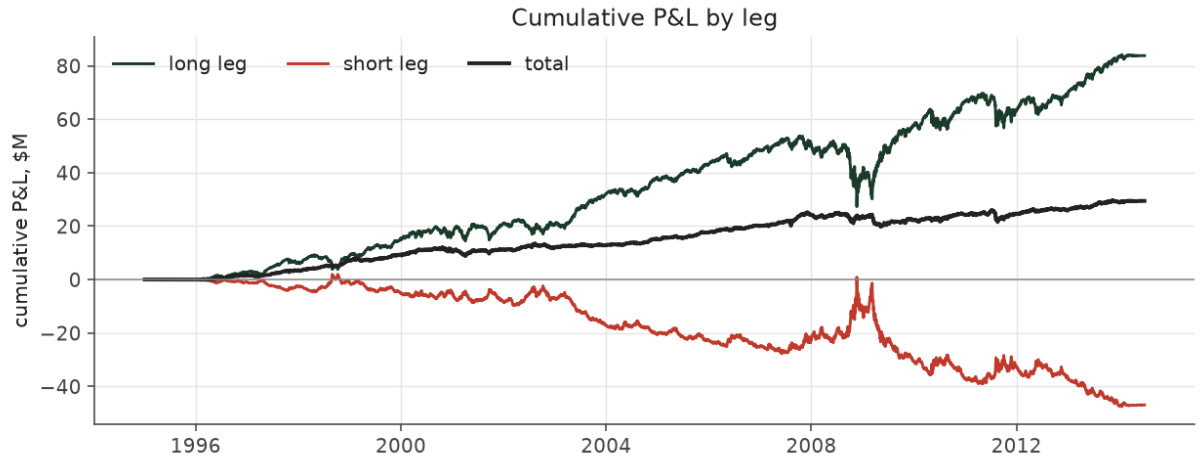


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	83.80	-46.97
P&L after costs ($M)	80.33	-50.91
of which common (average-event) component ($M)	28.93	-28.01

of which excess / PEAD component ($M)	14.01	16.42
excess minus costs ($M)	10.56	12.49
gross return per $ traded (bp)	453.74	-252.30
positions in the money (hit rate)	0.63	0.42
stand-alone Sharpe (leg only)	0.80	-0.48

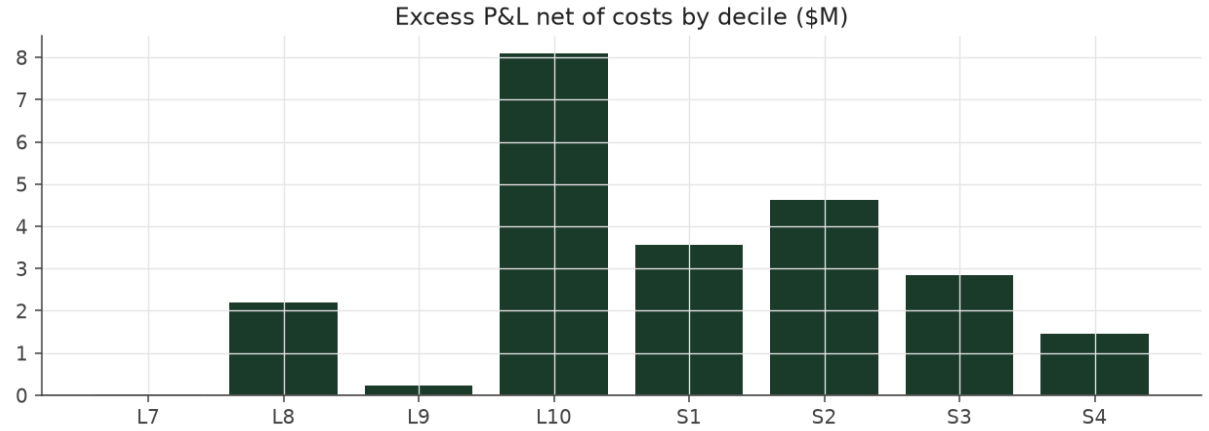
Daily P&L correlation between the two legs: **-0.97**.

• **The short leg loses money in absolute terms** (-47.0M before costs) while the long leg earns +83.8M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 28.9M / 28.0M. Stripping that common component, the surprise signal itself earned +14.0M on the long side and +16.4M on the short side.

• The two legs' daily P&L are -0.97 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.80 long / -0.48 short vs 1.11 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	151.48	6.63	0.31	0.29	0.02	0.1%	1.52
long	8	16,588	419.95	18.91	2.98	0.80	2.19	9.5%	52.10
long	9	17,747	579.02	19.40	1.30	1.07	0.23	1.0%	4.06
long	10	17,174	696.43	35.39	9.41	1.32	8.10	35.1%	116.27
short	1	17,169	687.39	-20.20	5.06	1.50	3.56	15.4%	51.76
short	2	17,281	581.80	-15.87	5.84	1.21	4.62	20.1%	79.49
short	3	15,970	427.74	-10.65	3.74	0.90	2.84	12.3%	66.46
short	4	9,171	164.64	-4.20	1.78	0.33	1.45	6.3%	87.95



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	331.20	4.24	6.87	1.26	5.61	24.3%	169.38
2	22,885	698.98	5.38	9.11	1.95	7.15	31.0%	102.34

3	25,501	855.45	6.19	7.67	1.78	5.89	25.6%	68.91
4	27,583	908.68	5.87	4.17	1.42	2.75	11.9%	30.29
5	28,947	914.14	7.73	2.61	1.00	1.61	7.0%	17.57

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	237.22	5.30	6.82	1.19	5.63	24.4%	237.28
2	23,084	737.56	4.92	8.94	2.21	6.73	29.2%	91.20
3	25,851	913.89	5.24	8.86	1.83	7.03	30.5%	76.92
4	27,665	916.54	5.74	3.53	1.28	2.25	9.8%	24.55
5	28,874	902.97	8.23	2.27	0.90	1.37	5.9%	15.18

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	1,393.00	-4.21	6.39	2.48	3.91	17.0%	28.09
41-60d	25,652	849.98	5.63	4.06	1.72	2.34	10.1%	27.50
61-90d	35,324	1,098.83	6.54	8.72	2.31	6.42	27.8%	58.40
>90d	12,298	366.64	21.45	11.26	0.91	10.35	44.9%	282.27

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	747.21	10.90	10.52	1.46	9.06	39.3%	121.23
2002-07	39,956	1,240.62	12.89	13.31	2.43	10.88	47.2%	87.72
2008-13	45,278	1,720.62	5.63	6.60	3.52	3.08	13.3%	17.88

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	361.52	6.68	2.59	0.76	1.83	7.9%	50.55
Chems	2,590	247.25	1.86	2.09	0.43	1.66	7.2%	67.09
Durbl	2,843	271.20	0.82	1.76	0.49	1.27	5.5%	46.95
Enrgy	6,194	306.46	2.68	5.00	0.58	4.42	19.2%	144.27
Hlth	11,696	351.27	4.78	2.81	0.74	2.07	9.0%	58.92
Manuf	12,034	328.49	2.00	2.19	0.68	1.51	6.6%	45.98
Money	17,583	324.19	2.90	3.99	0.73	3.26	14.2%	100.66
NoDur	5,081	297.45	3.35	2.27	0.56	1.71	7.4%	57.47
Other	18,377	351.79	2.42	2.88	0.78	2.10	9.1%	59.76
Shops	11,984	342.63	3.79	3.25	0.70	2.55	11.1%	74.40

Telcm	3,981	294.61	-1.67	0.91	0.54	0.37	1.6%	12.55
Utils	4,672	231.59	-0.19	0.69	0.42	0.26	1.1%	11.32

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	1,190.47	5.04	7.46	2.16	5.30	23.0%	44.53
low vol	40,091	1,262.78	13.29	12.94	2.76	10.18	44.1%	80.57
mid vol	37,903	1,199.44	10.75	9.88	2.38	7.50	32.6%	62.54

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	7.30%	0.00%
0.5%	1.9	7.02%	0.27%
1.0%	3.8	6.74%	0.56%
2.0%	7.6	6.12%	1.18%
5.0%	19.1	3.72%	3.58%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.